
Budget-Limited Failure, Not Factual Drift, Drives Error Scaling in Controlled Long-Context Summarization

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Abstract

Long-context language models are increasingly used to summarize records that are too long for routine human inspection. For these settings, the relevant question is not only whether summaries are faithful, but which error types grow as source length and fact load increase. We study this question with controlled long-document summarization tasks built from QMSUM and GOVREPORT background text. We inject known compliance-incident facts as ground truth, then ask GPT-4.1-MINI and GPT-5-MINI to return structured summaries across 48 main runs spanning roughly 2k, 6k, 12k, and 24k source tokens. The experiment includes fixed-load tasks with 10 target facts at every length and scaled-load tasks with 4, 8, 16, and 32 target facts. Across 600 target records and 536 returned records, hallucinations and duplicates are zero, normalized unit, object, count, month, and severity errors are zero, and only two action-field errors occur. The only strong length-dependent failure is structural: GPT-5-MINI returns empty visible content in two 24k-token, 32-fact runs under a 5,000 completion-token cap. Rerunning those cases with a 12,000-token cap recovers all records with zero omissions or field errors. These results suggest that, in this controlled setting, long-context summarization fails abruptly at a task-budget boundary rather than through gradual factual drift.

1 Introduction

Long-context language models are now used to summarize evidence that people cannot realistically read end to end: regulatory reports, meeting archives, legal records, and operational logs. In these settings, a single summary score is not enough. A system that almost never invents facts can still be unsafe if it omits low-salience incidents, changes counts, or fails to emit a usable output when the source is long.

What should fail as context grows? Our main question is simple: as source documents become longer, which summarization errors remain stable and which errors increase? Prior long-document work gives strong evidence that factuality and coherence need fine-grained evaluation Chang et al. [2023], Bishop et al. [2024]. Factuality benchmarks also show that entity, predicate, circumstance, and out-of-source errors behave differently Pagnoni et al. [2021], Tang et al. [2023]. However, these settings often mix source length, summary length, salience, and annotation difficulty. We focus on direct error scaling under controlled ground truth.

Our approach. We build controlled long-document tasks by inserting known compliance-incident records into natural QMSUM and GOVREPORT background documents Zhong et al. [2021], Huang et al. [2021]. Each injected record has a record id, unit, action, object, count, month, and severity. We then ask two real hosted models, GPT-4.1-MINI and GPT-5-MINI, to extract all incident records as strict JSON. The design has two task families: **FIXEDLOAD** holds the target fact count fixed at 10 while source length grows, and **SCALEDLOAD** increases both source length and target fact count.

Preview. The main result is negative but operationally useful. Across 48 main model calls, 600 target records, and 536 predicted records, hallucinations and duplicates are 0. Normalized unit, object, count, month, and severity errors are also 0, and only 2 action-field errors occur. The only strong length-dependent failure is not a gradual rise in factual substitutions. It is an abrupt visible-output failure: GPT-5-MINI returns empty content in two 24k-token, 32-fact runs because the full 5,000-token completion budget is consumed as reasoning tokens. A high-budget recovery rerun with 12,000 completion tokens fixes both cases with zero omissions and zero field errors.

We make three contributions:

- We propose a controlled long-context summarization design that preserves natural distractor text while giving exact record-level ground truth.
- We conduct a compact factorial experiment over two models, two task families, four source-length bins, and three replicates per condition.
- We show that factual error categories are nearly stable in valid outputs, while the apparent length effect comes from a completion-budget failure at the hardest scaled-load condition.
- We complement prior factuality benchmarks with a sensitivity analysis that separates parse failures from factual omissions.

Section 2 reviews long-document summarization and factuality evaluation. Section 3 describes the controlled task design and metrics. Section 4 presents the main results, recovery check, and statistical tests. Section 5 discusses implications and limitations.

2 Related Work

Long-document summarization. Long-document summarization benchmarks expose models to inputs whose structure and evidence density differ from short news articles. QMSUM introduces query-focused meeting summarization over multi-domain meeting transcripts Zhong et al. [2021], while GOVREPORT provides long government reports with expert summaries Huang et al. [2021]. These datasets motivate our choice of natural background text: we want realistic long-form distractors, but we also want exact ground truth for error typing.

Evaluation beyond aggregate scores. BoookScore studies book-length summarization and identifies long-summary coherence errors such as omissions, discontinuities, salience errors, and duplication Chang et al. [2023]. LongDocFACTScore adapts factuality scoring to long documents by retrieving evidence snippets before applying a factuality metric Bishop et al. [2024]. These works show that length creates evaluation challenges, but their main targets are summary quality and factual support. Our experiment instead holds the target facts under experimental control and estimates how specific error indicators change with length.

Factuality taxonomies and detectors. FRANK provides a fine-grained factuality taxonomy for abstractive summarization, including entity, predicate, circumstance, discourse, and out-of-source errors Pagnoni et al. [2021]. FactCC, SummaC, and MiniCheck develop automatic consistency or fact-checking models for source-grounded generation Kryscinski et al. [2020], Laban et al. [2022], Tang et al. [2024]. AggreFact shows that metric behavior varies across summarizer generations, datasets, and error types Tang et al. [2023]. Recent stress tests further show that factuality metrics can be sensitive to perturbations in long-document settings Mujahid et al. [2025]. We take a complementary route: instead of scoring free-form claims with a learned metric, we ask models to recover known structured records and score field-level mismatches directly.

Positioning. Our work is closest in motivation to studies that ask whether long-context summarization errors change qualitatively with length. The key difference is control. By injecting known facts into natural documents, we can distinguish omissions, hallucinated ids, duplicate ids, and field substitutions without human annotation. This makes the experiment small but diagnostic: it tests whether error categories slope upward under two controlled workloads, not whether a generated prose summary is globally good.

Family	Length bins	Target facts	Purpose
FIXEDLOAD	2k, 6k, 12k, 24k	10 at every length	Isolate context dilution with fixed summary workload.
SCALEDLOAD	2k, 6k, 12k, 24k	4, 8, 16, 32	Test growth in context length and fact workload together.

Table 1: Experimental design. Each model-family-length condition has three replicates, giving 48 main model calls across two models, two task families, four length bins, and three replicates.

3 Methodology

Task formulation. Each task document contains natural background text plus a set T of injected incident records. Each target record $t \in T$ has seven fields: record id, unit, action, object, count, month, and severity. A model returns a JSON object containing a predicted set P of records with the same schema. The scorer matches predictions to targets by record id, then evaluates omissions, hallucinated ids, duplicate ids, field mismatches, and exact-record recovery.

Datasets and task families. We use pre-downloaded QMSUM and GOVREPORT test-split text as natural background. The injected incidents are synthetic compliance notes, which gives exact ground truth while preserving long natural distractor context. Table 1 summarizes the two task families. FIXEDLOAD isolates context dilution by keeping 10 target records at every length. SCALEDLOAD tests a more realistic workload in which longer documents also contain more target facts.

Models and decoding. We query GPT-4.1-MINI and GPT-5-MINI through the OpenAI API after verifying that both models are available to the authenticated account. The main run uses JSON-mode chat completions with a 5,000 completion-token cap. We use deterministic decoding where supported by the model endpoint. After observing two high-load GPT-5-MINI failures, we run a recovery check on those exact tasks with `max_completion_tokens=12000`.

Metrics. Omission rate is the fraction of target ids missing from the returned JSON. Hallucination rate is the fraction of predicted ids not present in the source. Duplicate rate is the fraction of repeated predicted ids. Field-error rate is measured among matched records and checks unit, action, object, count, month, and severity. Exact-record rate is the fraction of target records whose id is returned and whose fields all match. We also record parse failure when the model output is empty or invalid JSON. The scorer normalizes harmless packaging differences, such as placing the count inside the object phrase when the separate count field is correct.

Baselines and statistical analysis. We use three sanity baselines. A perfect extractor has zero errors by construction and verifies the scoring logic. A prompt-format baseline records parse failures and duplicate ids, separating structural failures from factual errors. A cross-model comparison checks whether slopes are model-specific. For statistical tests, we fit binomial logistic models for each error indicator against $\log_2$ source tokens, adding model, task-family, or fact-count controls where possible. We report odds ratios per doubling of source length and Benjamini-Hochberg adjusted p -values.

4 Results

Main-run summary. Table 2 gives the aggregate experiment summary. The main run contains 48 model calls, 600 target records, and 536 predicted records. The mean per-run omission rate is 4.17%, while hallucination and duplicate rates are both 0. The mean field-error rate among matched records is 0.43%, driven by two action-field errors. All unit, object, count, month, and severity errors are 0 after normalization.

Valid outputs are nearly exact. The mean omission rate falls to 0 when parse failures are excluded. In those 46 valid runs, hallucination and duplicate rates remain 0, and the mean exact-record rate rises to 99.57%. This means the apparent omission growth in the main run is not caused by partial missed facts inside otherwise valid JSON. It comes from two empty visible outputs in the hardest condition.

The hardest condition separates the models. Table 3 focuses on the 24k-token bin. GPT-4.1-MINI returns all records in both fixed and scaled load, including all 96 records in the 32-fact scaled-load condition. GPT-5-MINI also returns all records in the 10-fact fixed-load condition. Its only failure is

Quantity	Value
Runs	48
Target records	600
Predicted records	536
Parse failures	2
Mean per-run omission rate	4.17%
Mean per-run hallucination rate	0.00%
Mean per-run duplicate rate	0.00%
Mean per-run field-error rate among matched records	0.43%
Mean per-run exact-record rate over target records	95.42%

Table 2: Overall main-run results. Rates are means over runs, matching the scoring summary. Hallucination and duplicate rates are best possible and shown in **bold**.

Model	Family	Facts	Parse failures	Omission	Hallucination	Duplicate	Exact record
GPT-4.1-MINI	FIXEDLOAD	10	0/3	0.0%	0.0%	0.0%	100.0%
GPT-4.1-MINI	SCALEDLOAD	32	0/3	0.0%	0.0%	0.0%	100.0%
GPT-5-MINI	FIXEDLOAD	10	0/3	0.0%	0.0%	0.0%	100.0%
GPT-5-MINI	SCALEDLOAD	32	2/3	66.7%	0.0%	0.0%	33.3%

Table 3: Performance in the 24k-token condition. The only large error rate appears for GPT-5-MINI under scaled load, where two of three responses contain no visible JSON because the main-run completion budget is exhausted. Best values are in **bold**.

the 32-fact scaled-load condition: two of three runs terminate by length with empty visible content, producing 64 missing records and a mean omission rate of 66.7%.

A larger completion budget recovers the failures. Table 4 shows the high-budget recovery check. The two failed GPT-5-MINI 24k-token, 32-fact tasks recover completely with `max_completion_tokens=12000`. Both runs stop normally, return all 32 records, and have zero hallucinations, duplicates, and field errors. The recovered runs use 2,752 and 2,560 reasoning tokens, respectively, showing that the main-run failures were budget-sensitive rather than evidence-retrieval failures.

Length slopes reflect separation at one corner case. Table 5 reports logistic slope tests. Omission grows in the main run under pooled and scaled-load analyses, with Benjamini-Hochberg adjusted $p = 0.0040$. Exact-record error also grows in scaled load. However, the odds ratios are infinite because all omission events occur at the largest scaled-load bin. Field errors remain approximately stable: the fixed-load action-error odds ratio is 1.09 per doubling of source tokens with adjusted $p = 0.8703$.

5 Discussion

The main failure mode is abrupt, not gradual. The experiment was designed to find which factual errors grow with length. In this controlled setting, most did not grow. Hallucinated ids, duplicate ids, unit errors, count errors, month errors, severity errors, and object errors are all zero in the main run. The only observed field substitutions are two action errors. The strongest length effect is instead a structural failure: GPT-5-MINI spends the 5,000-token completion budget before emitting visible JSON in two high-load runs.

Output budget is part of long-context reliability. This matters because users often experience an empty or invalid output as a summarization failure, even when the model may have internally identified the needed facts. The recovery check shows that the two failed tasks are solvable by the same model when the completion budget is raised. For long-context systems, source length, target fact count, output schema size, and hidden reasoning-token allocation are therefore coupled. Evaluation should report these quantities alongside factuality metrics.

The result does not imply long summaries are solved. Our injected facts are synthetic and intentionally identifiable as compliance incident notes. Real summarization often requires deciding

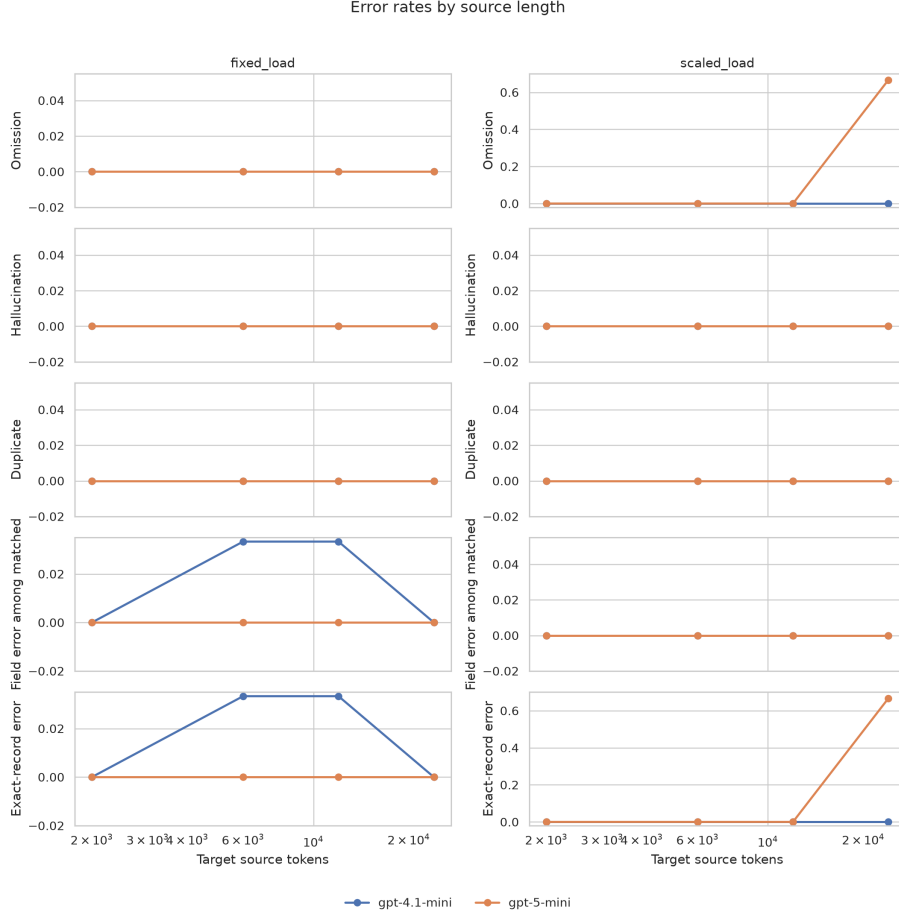


Figure 1: Error rates by source length for each model and task family. Most factual error categories remain at zero across length. The visible increase in omission and exact-record error is concentrated in the GPT-5-MINI SCALEDLOAD 24k-token condition, where two runs return empty visible content under the 5,000-token completion cap.

Task	Target facts	Parse failure	Missing	Field errors	Reasoning tokens	Completion tokens
scaled_load-24000-0	32	0	0	0	2,752	4,900
scaled_load-24000-1	32	0	0	0	2,560	4,706

Table 4: High-budget recovery for the two failed GPT-5-MINI scaled-load 24k runs. Increasing the completion cap from 5,000 to 12,000 tokens recovers every record with zero missing records and zero field errors. Best values are in **bold**.

salience, resolving implicit references, and synthesizing events across sentences. The JSON schema also constrains outputs tightly, which likely suppresses hallucinations and duplicates. A free-form prose summary could show more relation errors, discourse errors, or salience-dependent omissions.

Scope and limitations. The study uses two OpenAI models, 48 main calls, three replicates per condition, and a maximum source length of roughly 24k tokens. This is long enough to exceed routine human quick-reading workflows, but far below million-token contexts. The main omission result also depends on treating parse failures as omissions. That is appropriate for an end-user extraction system, but it is not the same mechanism as retrieving a fact and then choosing not to mention it. Finally, the controlled ground truth avoids human annotation cost, but it does not evaluate prose quality, narrative coherence, or whether the selected facts would be salient to a human summarizer.

Implications. The practical lesson is to distinguish factual drift from budget failure. If a system is evaluated only by aggregate omission rate, the 24k-token GPT-5-MINI condition looks like a

Outcome	Scope	Odds ratio per 2x tokens	BH-adjusted p	Interpretation
Omission	Pooled adjusted	∞	0.0040	Grows in the main run.
Omission	Scaled load	∞	0.0040	Grows in the main run.
Exact-record error	Scaled load	∞	0.0040	Grows in the main run.
Any field error	Fixed load	1.09	0.8703	Approximately stable.
Action error	Fixed load	1.09	0.8703	Approximately stable.

Table 5: Logistic length-slope tests. Infinite odds ratios indicate separation: all omission events occur in the largest scaled-load bin. Significant values are in **bold**.

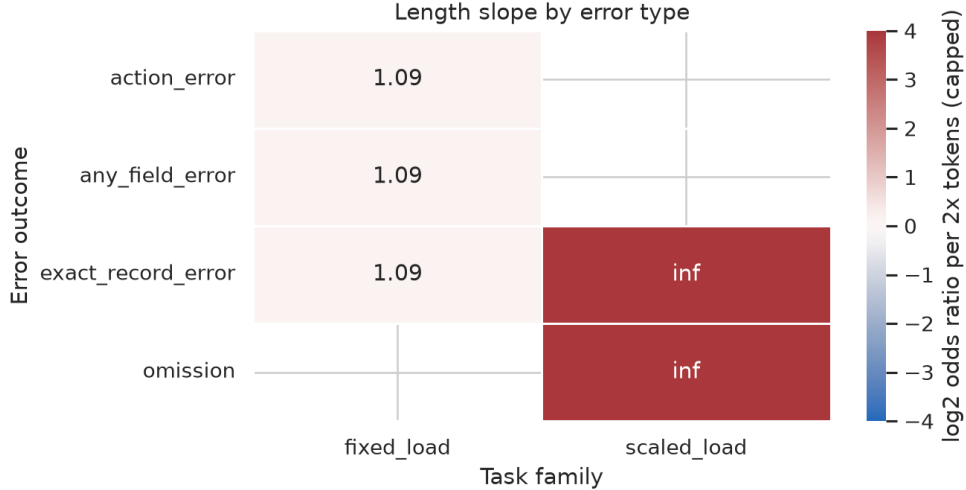


Figure 2: Length-slope heatmap for tested error indicators. Significant main-run slopes appear for omissions and scaled-load exact-record errors, but these effects are separation artifacts caused by two empty GPT-5-MINI outputs at the largest scaled-load bin. Field-error slopes are small and statistically indistinguishable from zero.

severe length-scaling failure. The recovery check changes the interpretation: the underlying task remains recoverable, but the original configuration is underspecified for the combined context and output workload. Budget sweeps and reasoning-effort controls should be standard in long-context factuality evaluations.

6 Conclusion

We study how error types change as long-context summarization tasks grow in source length and target fact load. Using natural QMSUM and GOVREPORT background text with injected compliance-incident ground truth, we score 48 main runs from GPT-4.1-MINI and GPT-5-MINI at record and field level. The key finding is that ordinary factual errors are nearly stable in valid outputs: hallucinations and duplicates are zero, most field-error categories are zero, and only two action-field errors appear.

The only strong length-dependent failure is an output-budget failure for GPT-5-MINI at 24k source tokens and 32 target facts. Raising the completion-token cap recovers both failed runs with zero omissions and zero field errors. Future work should sweep completion budgets, reasoning-effort settings, larger contexts, and less explicit naturally occurring facts. The central takeaway is that long-context summarization reliability should be evaluated by error mechanism, not only by aggregate error rate.

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